Exploring the Gender Wage Gap

Emily Morra [1] & Samantha Miller [2]

[1] Affiliation 1; emorra@ramapo.edu
[2] Affiliation 2; smille20@ramapo.edu

Abstract: For this study, the questions addressed were as follows: Is the gender wage gap closing? Are there additional factors, if any, that are also contributing to the gap in salaries? To answer these questions, the data sets used explored several different factors in addition to gender, such as employment type, education level, ethnicity, etc. By assessing all these components, we were able to determine whether they were somehow connected to the focus: salary. We used exploratory data analysis and made use of visualizations such as histograms, line graphs, and box plots as well as multiple linear regression and testing versus training models. After examining all this data, it was clear to see that there is a disparity in pay between males and females. Also, ethnicity seemingly plays a large role in this issue as well. There were not many clear-cut trends that could help determine whether the gender wage gap is decreasing, but we can confirm that it's still an extremely prevalent issue across multiple industries.

Keywords: gender wage gap; ethnicity; disparity

1. Introduction

2. Materials and Methods

For our study, we utilized exploratory data analysis. First, we compared the number of males and females in each data set to make sure that we'd be examining an evenly spread dataset. The only dataset that was extremely uneven was the IT Specialists set, but this can speak to the disparity in the number of males versus females in the tech industry. We made use of visualizations such as box plots, bar plots, line graphs, and histograms, and included multiple linear regression models as well as training and testing comparisons.

3. Results - Below are the findings for each of the four data sets we explored.

3.1.1. San Francisco Data Set

3.1.1. Male & Female Totals

- Total Females: 51,349
- Total Males: 67,820

3.1.1. Means & Maximums

- Mean total pay for females: $68,448.29
- Mean total pay for males: $83,923.70
- Maximum total pay for females: $390,111.98
- Maximum total pay for males: $567,595.43

3.1.2. Texas Data Set

3.1.1. Male & Female Totals

- Total Females: 84,818

- Total Males: 63,254

3.1.2. Multiple Linear Regression

- R-squared: 0.02205

3.1.2. Training vs Testing

- Training R-squared: 0.1955896

3.1.3. IT Specialists Data Set

3.1.1. Male & Female Totals

- Total Females: 60
- Total Males: 158

3.1.3. Multiple Linear Regression

- R-squared: 0.02776

3.1.3 Training vs Testing

- Training R-squared: 0.1022302

3.1.4 White House Data Set

3.1.1. Male & Female Totals

- Total Females: 5,577
- Total Males: 4,978

3.1.4. Means, Medians, & Maximum

- Mean total pay for females: $68,641.35
- Mean total pay for males: $83,051.63
- Median total pay for females: $55,717
- Median total pay for males: $72,700
- Maximum pay overall: $239,595 (male)

3.1.4 Multiple Linear Regression

- R-squared: 0.1613

3.1.4 Training vs Testing

- Training R-squared: 0.06763142

3.2. Figures and Tables

Table 1 – Fit model for all predictors in IT Specialists data set

term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
`Years of Experience`	-56321.424	30874.67	-1.8241952	0.070181332
GenderMale	-88436.317	116271.51	-0.7606018	0.448130010
Age	-5164.988	10016.75	-0.5156352	0.606894499
EducationMaster's	-168071.161	130034.18	-1.2925152	0.198234661
EducationNone	83234.451	120145.99	0.6927776	0.489557205
`Employment Type`Full-time	-57991.405	175372.95	-0.3306747	0.741367523
`Employment Type`Part-time	-20958.162	171465.50	-0.1222296	0.902886415
`Employment Type`Remote	115811.321	233608.26	0.4957501	0.620821273
`Number of Projects`	3015.523	11822.55	0.2550653	0.799034065
`Job Satisfaction`	115185.789	77113.41	1.4937193	0.137421892

Table 2 – Predictions for training data for IT Specialists data set

.pred <dbl>	Annual Salary <dbl>	Gender <chr>
1156068	395436.0	Male
1703935	2334960.0	Male
1005644	1392960.0	Male
1407467	1338795.0	Female
1576237	1752935.0	Female
1459384	1687111.0	Female
1344079	902935.0	Male
1267559	727111.0	Male
1180468	732821.0	Male
1337081	1630396.0	Male

Table 3 – Percentage of variability explained by training model of IT Specialists data set

.metric <chr>	.estimator <chr>	.estimate <dbl>
rsq	standard	0.1022302

1 row

Table 4 – Prediction for testing data for IT Specialists data set

.pred <dbl>	Annual Salary <dbl>	Gender <chr>
1219199	1507343.0	Male
1347136	1124939.0	Male
1279960	1630396.0	Male
1203086	1009436.0	Male
1121469	758376.0	Male
1022229	1556575.0	Male
1317186	1541572.0	Female
1180646	1205874.0	Male
1267559	727111.0	Male
1197600	495252.0	Male

Table 5 – Percentage of variability explained by testing model for IT Specialists data set

.metric <chr>	.estimator <chr>	.estimate <dbl>
rsq	standard	0.01414929

Table 6 – Fit model for all predictors in White House data set

term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>
(Intercept)	96020.47	1754.7871	54.719156
genderMale	14295.12	866.8223	16.491410
statusEmployee	-29342.49	1758.8997	-16.682300
statusEmployee (part-time)	-46020.47	38525.8693	-1.194534
pay_basisPer Diem	-76824.82	19249.6875	-3.990964

Table 7 – Predictions for training data for White House data set

.pred <dbl>	salary <int>	gender <chr>
80973.10	130500	Male
66677.98	28000	Female
66677.98	115000	Female
66677.98	75000	Female
80973.10	28000	Male
80973.10	130500	Male
66677.98	72022	Female
80973.10	157000	Male
66677.98	100904	Female
80973.10	36000	Male

Table 8 – Percentage of variability explained by training model for White House data set

.metric <chr>	.estimator <chr>	.estimate <dbl>
rsq	standard	0.06763142

Table 9 – Prediction for testing data for White House data set

.pred <dbl>	salary <int>	gender <chr>
80973.10	25000	Male
66677.98	22000	Female
66677.98	27964	Female
80973.10	30000	Male
96020.47	108824	Female
66677.98	25000	Female
80973.10	30000	Male
66677.98	25897	Female
66677.98	33792	Female
80973.10	100000	Male

Table 10 – Percentage of variability explained by testing model for White House data set

.metric <chr>	.estimator <chr>	.estimate <dbl>
rsq	standard	0.05496961

Table 11 – Fit model for all predictors in Texas data set

term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>
(Intercept)	48583.589	984.2239	49.3623325
ETHNICITYASIAN	11276.200	1059.8578	10.6393519
ETHNICITYBLACK	-7205.571	991.9756	-7.2638597
ETHNICITYHISPANIC	-3925.639	990.8483	-3.9618965
ETHNICITYOTHER	-3130.681	1385.6375	-2.2593794
ETHNICITYWHITE	3496.092	987.5409	3.5402002
GENDERMALE	5090.951	136.7767	37.2208943
STATUSCRP - CLASSIFIED REGULAR PART-TIME	-26370.300	621.4877	-42.4309282
STATUSCTF - CLASSIFIED TEMPORARY FULL-TIME	-10222.169	1886.1918	-5.4194751
STATUSCTP - CLASSIFIED TEMPORARY FULL-TIME	-28159.842	22380.3450	-1.2582398

Table 12 – Predictions for training data for Texas data set

.pred <dbl>	ANNUAL <dbl>	GENDER <chr>
46468.97	28903.08	MALE
44657.95	56515.80	FEMALE
52079.68	63000.00	FEMALE
44657.95	34737.84	FEMALE
59585.47	30000.00	FEMALE
59859.79	93480.00	FEMALE
49748.90	44642.04	MALE
57170.63	40641.96	MALE
52079.68	79892.64	FEMALE
57170.63	95191.92	MALE

Table 13 – Percentage of variability explained by training model of Texas data set

.metric <chr>	.estimator <chr>	.estimate <dbl>
rsq	standard	0.1955896

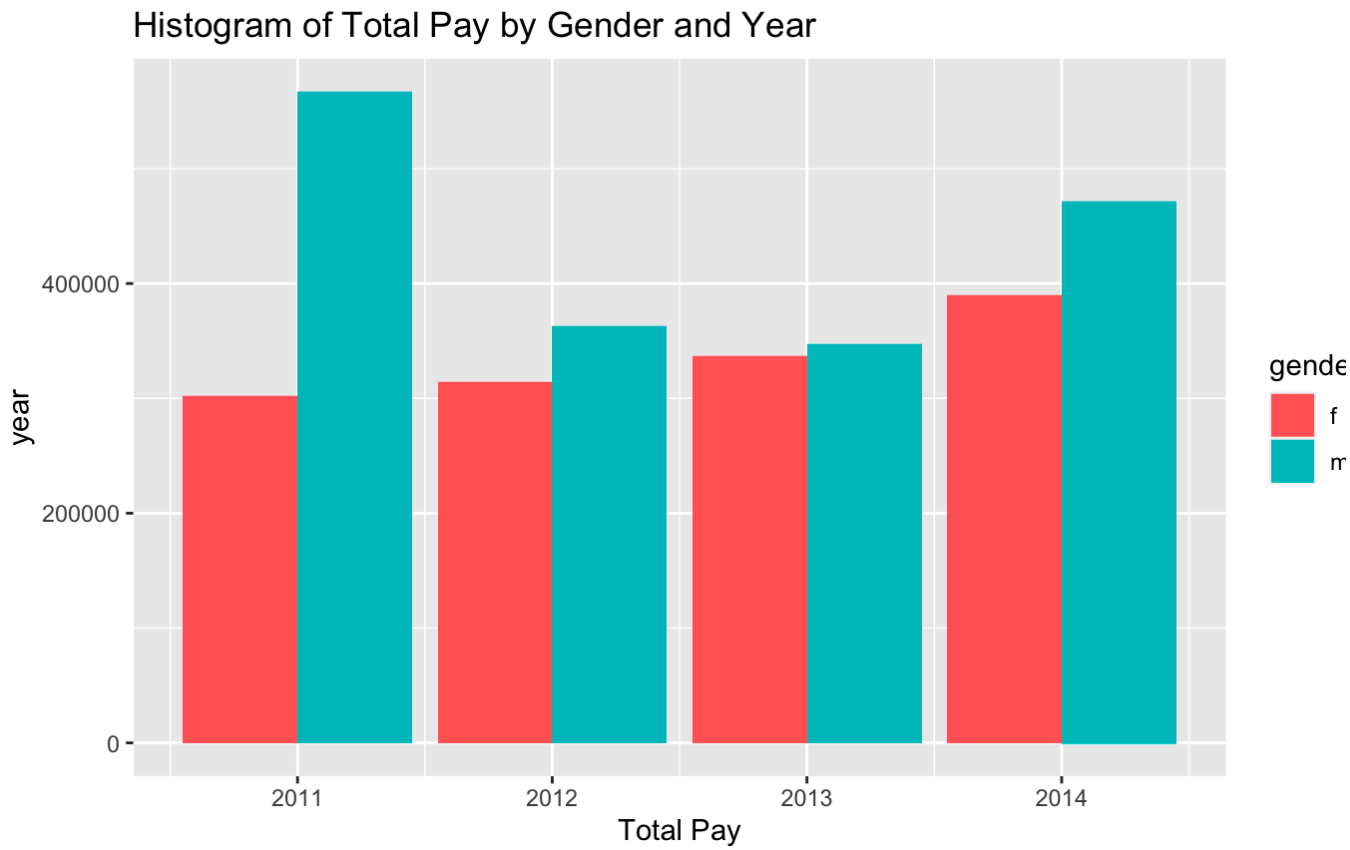
Table 14 – Prediction for testing data for Texas data set

.pred <dbl>	ANNUAL <dbl>	GENDER <chr>
41378.02	39411.24	FEMALE
44657.95	55129.92	FEMALE
46468.97	47400.00	MALE
30800.33	19944.00	MALE
30800.33	9018.00	MALE
63458.20	28236.96	FEMALE
67007.20	96000.00	FEMALE
67007.20	102000.00	FEMALE
56831.43	60000.00	FEMALE
56831.43	30000.00	FEMALE

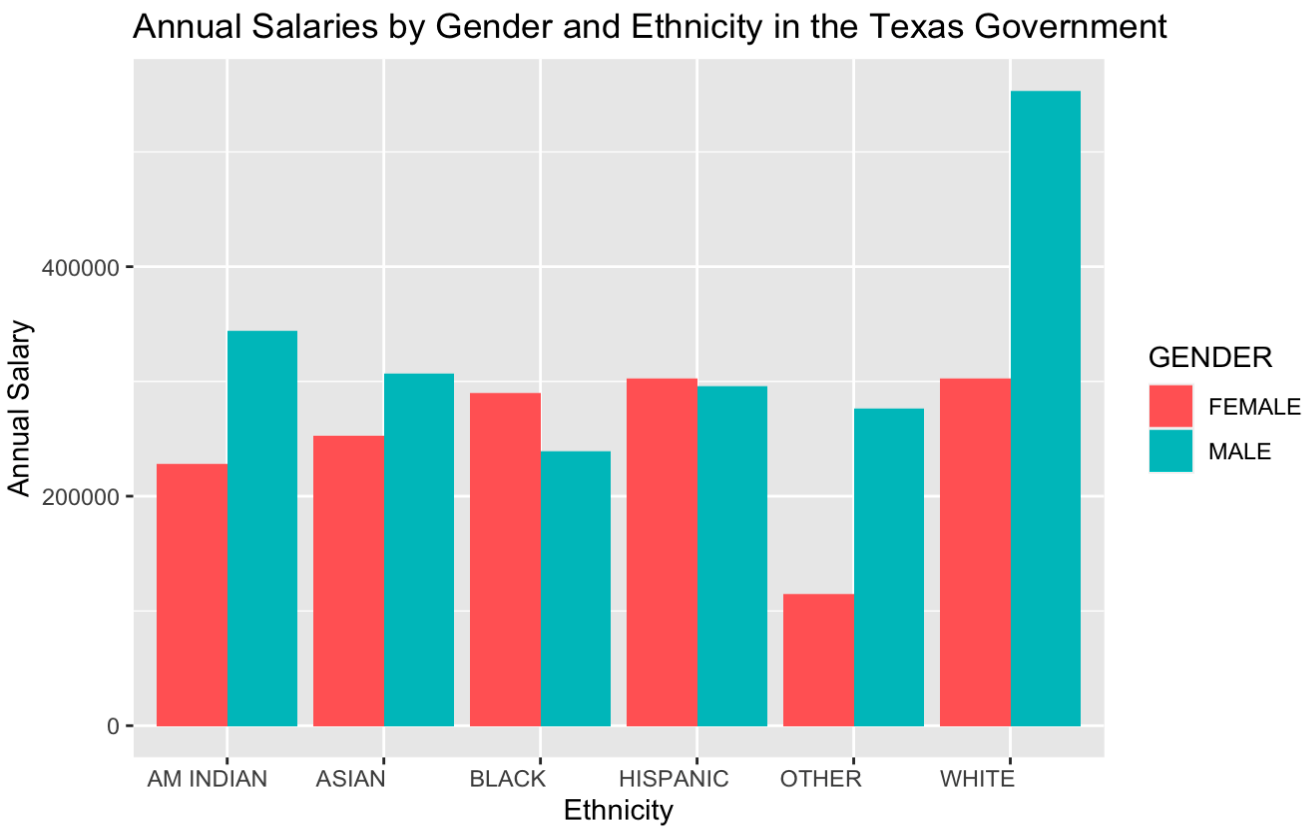
Table 15 – Percentage of variability explained by testing model for Texas data set

.metric <chr>	.estimator <chr>	.estimate <dbl>
rsq	standard	0.1789842

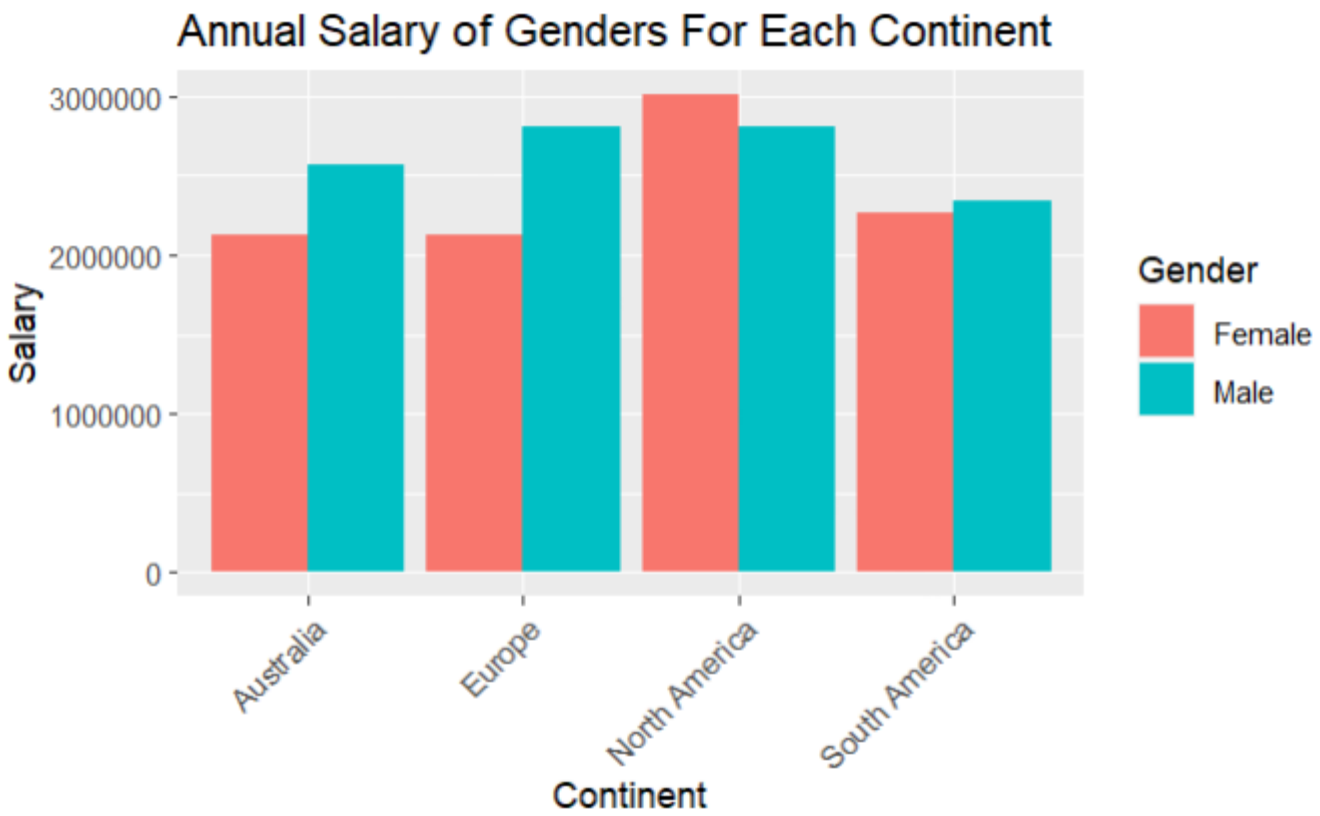
(a) Box plot comparing male and female salary from San Francisco data set.



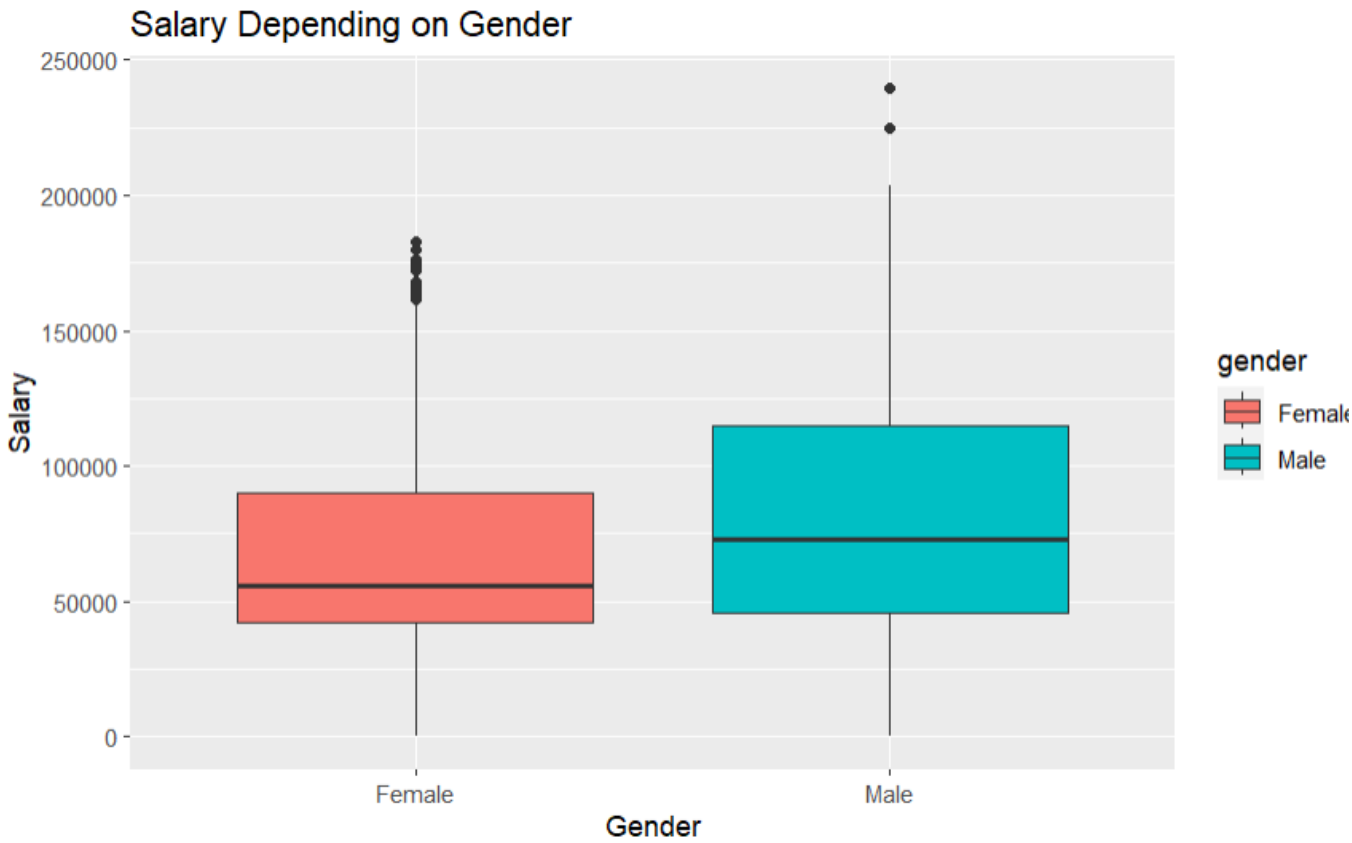
(**b**) Histogram comparing annual salary for males and females for each year from San Francisco data set.



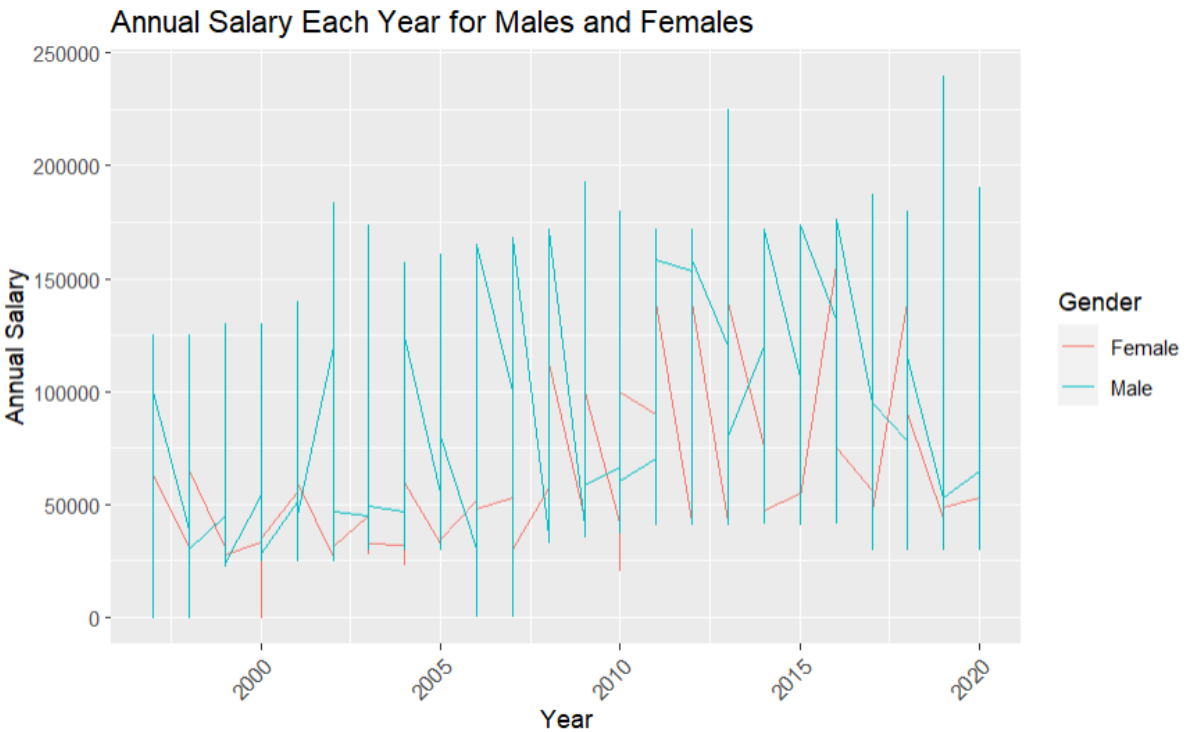
(**c**) Histogram comparing annual salaries of males and females for different ethnicities from Texas data set.



(**d**) Histogram comparing annual salary for males and females for each continent from IT Specialists data set.



(**e**) Box plot comparing male and female salary from White House data set.



(**f**) Line graph comparing annual salary for males and females for each year from White House data set.

4. Discussion

Based on these results, it's clear to see that the gender wage gap is still prevalent. In Figures a-f, there is a clear trend that male salaries are either greater than or equal to those of females for most of the data. In Tables 1-5 and 11-15, the training data has a higher R-squared value than that of the original multiple linear regression, and that the RMSE is lower for the training data. However, in Tables 6-10, we can observe that the opposite is true: the R-squared values in both the testing and training models are lower than that of the original multiple linear regression model, perhaps due to multicollinearity.

5. Conclusions

Although it is difficult to observe if the gender wage gap is closing, it's still clear to see that there are plenty of strong disparities between male and female pay overall. In addition to gender contributing to pay gaps, ethnicity, specifically for white males, was found to also play a significant role in this difference.

Data Availability Statement: All datasets were taken from Kaggle.

References

Adiquant. "White House Staff Salaries [2017-2020]." *Kaggle*, 6 July 2020, www.kaggle.com/datasets/adiquant/white-house-staff-salaries-20172020?rvi=1.

daniell_pr. "Salaries of It Specialists." *Kaggle*, 25 Aug. 2023, www.kaggle.com/datasets/daniell201254/salaries-of-it-specialists.

Ofer, Dan. "Texas Government Salaries." *Kaggle*, 21 May 2020, www.kaggle.com/datasets/danofer/texas-government-salaries?rvi=1.

Troncoso, Ronald. "SF Salaries (Gender Column Included)." *Kaggle*, 11 Dec. 2016, www.kaggle.com/datasets/ronaldtroncoso20/sf-salaries-extended.